Supporting Information for

# Switching Rules and Mismatch Dynamics in Context-Dependent Encoding

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This PDF file includes:

Supporting text

Figures S1 to S2

**Supporting Information Text**

Belief Update

Let $\pi_t(c) = P(C_t = c \mid R_{1:t})$ denote the posterior belief over contexts at time $t$. Belief was updated recursively by first forming a predictive prior from the previous posterior and the context transition matrix,

$$\pi_t'(c) = \sum_{c'} P(C_t = c \mid C_{t-1} = c')\, \pi_{t-1}(c'),$$

Here, the predictive prior $\pi_t'(c)$denotes the context belief before incorporating the current response $R_t$. This prior was combined with the response likelihood:

$$\pi_t(c) = \frac{p(R_t \mid c)\, \pi_t'(c)}{\sum_{c''} p(R_t \mid c'')\, \pi_t'(c'')}.$$

The inferred context was taken to be the maximum a posteriori (MAP) estimate of the posterior belief over contexts. Here $p(R_t \mid c)$ denotes the predictive response density under context $c$, computed numerically from that context's stimulus distribution and the encoding function in use at that time step.

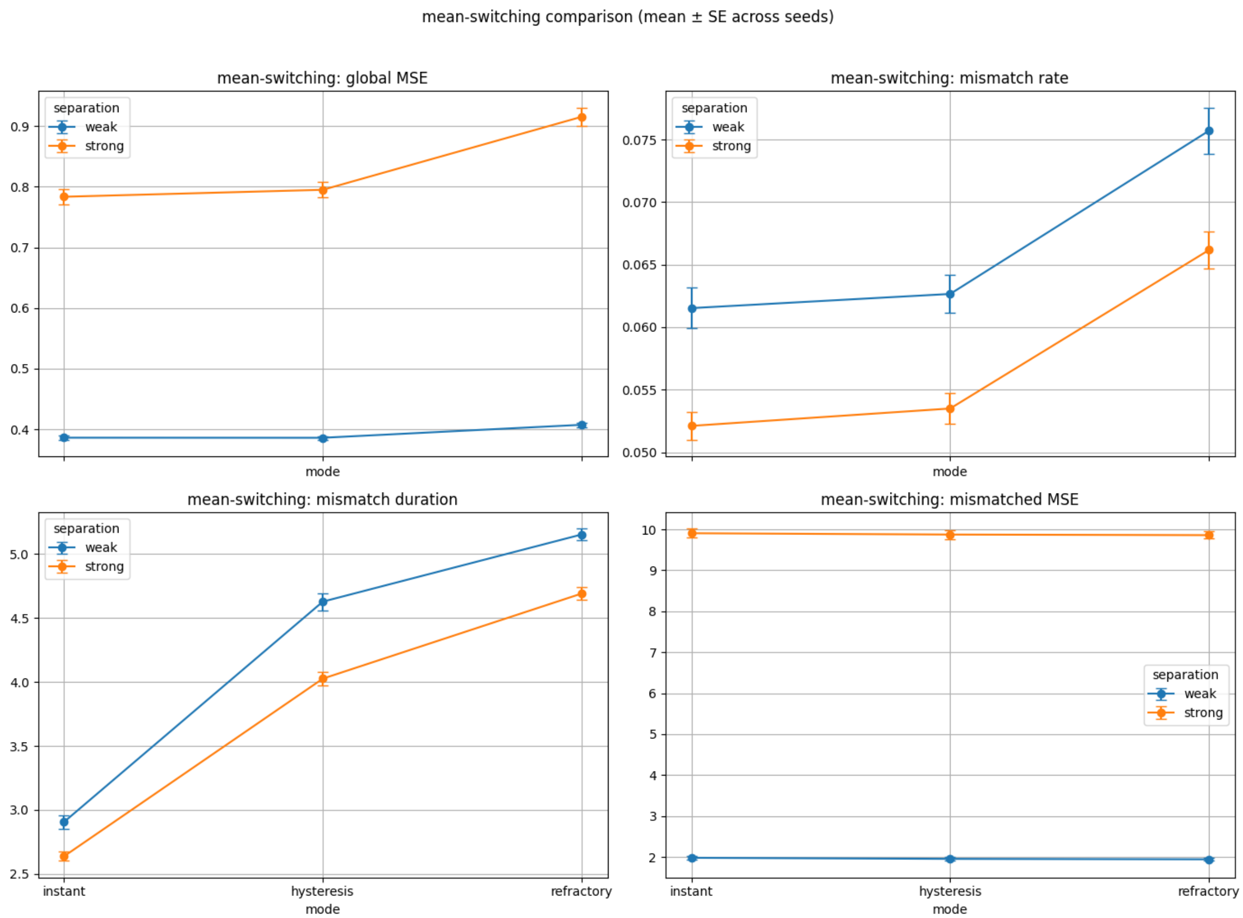


**Figure S1**. Context separation in the mean-switching environment trades improved inference against higher mismatch cost. Weak and strong separation correspond to $(\mu_0, \mu_1) = (-1,1)$ and $(\mu_0, \mu_1) = (-2,2)$, respectively, with $\sigma = 1$. Curves compare weak and strong separation across switching rules for global MSE, context misclassification rate, mean mismatch duration, and mismatched MSE (mean ± SE across seeds). Stronger separation reduces misclassification and shortens mismatch episodes, but it also increases mismatched MSE. As a result, global MSE is higher under strong separation for all three switching rules.

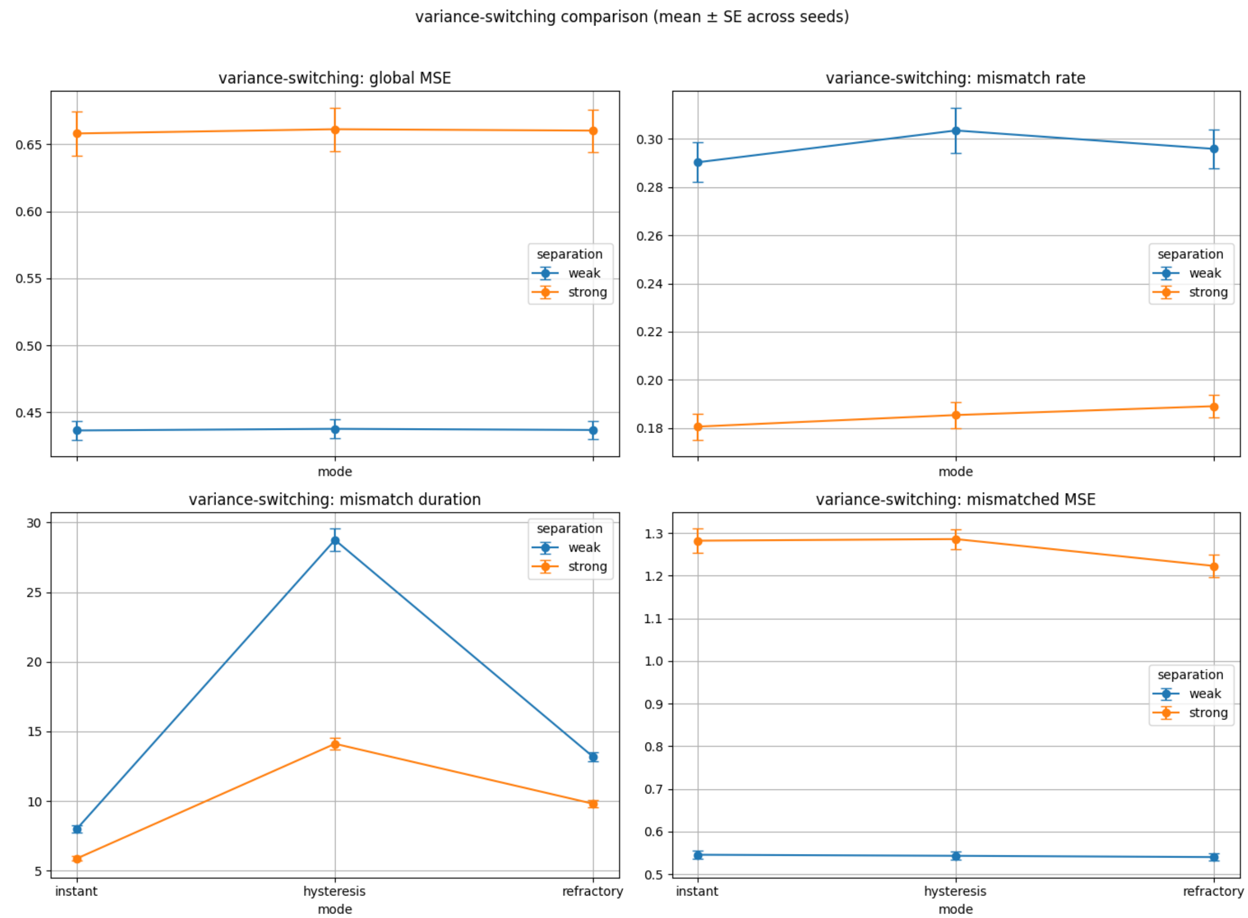


**Figure S2**. Context separation in the variance-switching environment trades improved inference against higher mismatch cost. Weak and strong separation correspond to $(\sigma_0, \sigma_1) = (0.9, 1.4)$ and $(\sigma_0, \sigma_1) = (0.7, 1.8)$, respectively, with $\mu = 0$. Curves compare weak and strong separation across switching rules for global MSE, context misclassification rate, mean mismatch duration, and mismatched MSE (mean ± SE across seeds). Stronger separation reduces context misclassification and mismatch duration but increases mismatched MSE. As a result, global MSE is higher under strong separation for all three switching rules, although the increase is smaller than in the mean-switching environment and the dependence on switching rule remains weak.